

Circles and circle theorems - Set 1 - Solutions

Specification MM3.2, MM3.3, M5.8, M5.9.

Answer key

Question	Answer	Topic
1	B	Circles and circle theorems
2	B	Circles and circle theorems
3	D	Circles and circle theorems
4	D	Circles and circle theorems
5	C	Circles and circle theorems
6	F	Circles and circle theorems
7	B	Circles and circle theorems
8	D	Circles and circle theorems
9	B	Circles and circle theorems
10	E	Circles and circle theorems
11	E	Circles and circle theorems
12	A	Circles and circle theorems

Worked solutions

Question 1 CIR-01 • Circles and circle theorems • spec MM3.2

Find the radius of the circle $x^2 + y^2 - 6x + 4y - 12 = 0$.

Answer: B

$$(x - 3)^2 + (y + 2)^2 = 12 + 9 + 4 = 25, \text{ so } r = 5.$$

Question 2 CIR-03 • Circles and circle theorems • spec MM3.2

The circle with centre $(2, -1)$ passes through $(5, 3)$. Find the equation of the tangent to the circle at $(5, 3)$.

Answer: B

Radius direction $(3, 4)$ has gradient $\frac{4}{3}$, so the tangent has gradient $-\frac{3}{4}$:
 $y - 3 = -\frac{3}{4}(x - 5) \Rightarrow 4y - 12 = -3x + 15 \Rightarrow 3x + 4y = 27.$

Question 3 CIR-04 • Circles and circle theorems • spec MM3.2

The points $(1, 7)$ and $(5, -1)$ are the ends of a diameter of a circle. Find the equation of the circle.

Answer: D

$$\text{Centre is the midpoint } (3, 3); \text{ radius}^2 = (3 - 1)^2 + (3 - 7)^2 = 4 + 16 = 20.$$

Question 4 CIR-06 • Circles and circle theorems • spec MM3.3

A, B and C lie on a circle with centre O . Angle $AOB = 100^\circ$ and C lies on the major arc AB . Find angle ACB .

Answer: D

The angle at the centre is twice the angle at the circumference subtended by the same arc: $ACB = 50^\circ$.

Question 5 CIR-07 • Circles and circle theorems • spec MM3.3

$PQRS$ is a cyclic quadrilateral with angle $P = 3x$, angle $R = x + 40^\circ$. Find x .

Answer: C

Opposite angles of a cyclic quadrilateral sum to 180° : $4x + 40 = 180, x = 35^\circ$.

Question 6 CIR-08 • Circles and circle theorems • spec MM3.3

A chord of a circle of radius 13 is at perpendicular distance 5 from the centre. Find the length of the chord.

Answer: F

The perpendicular from the centre bisects the chord: half-length $= \sqrt{13^2 - 5^2} = 12$, so the chord is 24.

Question 7 CIR-10 • Circles and circle theorems • spec MM3.3

TA is a tangent to a circle at A , and AB is a chord. C is a point on the circle on the opposite side of AB from T , with angle $ACB = 65^\circ$. Find angle TAB .

Answer: B

Alternate segment theorem: the angle between tangent and chord equals the angle in the alternate segment, so $TAB = 65^\circ$.

Question 8 CIR-12 · Circles and circle theorems · spec MM3.2

The circle C has equation $x^2 + y^2 - 4x - 6y + 9 = 0$. Which of the following lines does not meet C ?

Answer: D

Centre $(2, 3)$, radius 2, so the circle spans $0 \leq x \leq 4$ and $1 \leq y \leq 5$. The line $y = 0$ is 3 from the centre, more than the radius, so it misses; $x = 0$, $x = 4$ and $y = 5$ touch and $x = 2$ passes through the centre.

Question 9 CIR-13 · Circles and circle theorems · spec MM3.2

The circle $x^2 + y^2 = r^2$ and the line $3x + 4y = 20$ have exactly one point in common. Find r .

Answer: B

Distance from origin to the line $= \frac{20}{\sqrt{9+16}} = 4$. Tangency means $r = 4$.

Question 10 CIR-14 · Circles and circle theorems · spec MM3.3

AB is a diameter of a circle and C is a point on the circle with $AC = 6$ and $BC = 8$. Find the radius of the circle.

Answer: E

The angle in a semicircle is a right angle, so $AB = \sqrt{36 + 64} = 10$ and the radius is 5.

Question 11 CIR-15 · Circles and circle theorems · spec MM3.2

A circle passes through $(0, 0)$, $(6, 0)$ and $(0, 8)$. Find the coordinates of its centre.

Answer: E

The triangle has a right angle at the origin, so $(6, 0)$ – $(0, 8)$ is a diameter; the centre is its midpoint $(3, 4)$.

Question 12 X2-10 · Circles and circle theorems · spec MM3.2

The circle C has centre (a, b) and radius r . P : ' $r > |b|$ '. Q : ' C crosses the x -axis at two distinct points'. Which one of the following is true?

Answer: A

The distance from the centre to the x -axis is $|b|$; the circle cuts the axis twice exactly when this is less than r .